

Jeho Lee

Ph.D. Candidate (expected graduation: February 2027)
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RESEARCH INTERESTS

Hardware-aware on-device AI systems, with a focus on mobile NPU inference optimization and heterogeneous accelerator scheduling.

EDUCATION

Yonsei University, Seoul, Republic of Korea Mar 2020 – Present
 Ph.D. Candidate in Computer Science and Engineering
 Advisor: Hojung Cha | Expected graduation: February 2027

Ajou University, Suwon, Republic of Korea Mar 2015 – Feb 2020
 B.S. in Computer Engineering | GPA: 4.0/4.5

AWARDS AND HONORS

Best Paper Award, ACM MobiSys 2025 (2 of 233 submissions) Jun 2025
Rising Star, ACM MobiSys 2025 Jun 2025
SIGMOBILE Student Travel Grant, ACM MobiSys 2025 May 2025

PUBLICATIONS

* Equal contribution. Titles link to the publication when available.

Conference and Journal Papers

- [P1] DynaStitch: Dynamic ViT Inference via Neural Stitching for Resource-Constrained Edge Devices
 Jiwon Kim, Luca Mottola, **Jeho Lee**, Seonghoon Park, Hojung Cha, Thiemo Voigt
 ACM International Conference on Embedded Software (**EMSOFT 2026**), accepted.
- [P2] viNPU: Optimizing Vision Transformer Inference on Mobile NPUs
Jeho Lee, Gunjoong Kim, Chanyoung Jung, Jaehee Kim, Seonghoon Park, Hojung Cha
 European Conference on Computer Systems (**EuroSys 2026**).
- [P3] Phoenix: Thermal-Aware On-Device Inference of Multi-Instance DNNs for Mobile Video Applications
 Seunghyeok Jeon, Jiwon Kim, **Jeho Lee**, Hojung Cha
 ACM Transactions on Embedded Computing Systems (**TECS**), 25(2), Article 31, 2026.
- [P4] Vega: Fully Immersive Mobile Volumetric Video Streaming with 3D Gaussian Splatting
 Gunjoong Kim*, Seonghoon Park*, **Jeho Lee**, Chanyoung Jung, Hyungchol Jun, Hojung Cha
 ACM International Conference on Mobile Computing and Networking (**MobiCom 2025**).
- [P5] EOS: Energy-Optimized Super-Resolution on Mobile Devices for Live 360-Degree Videos
 Seonghoon Park, Minchan Kim, Hyejin Park, **Jeho Lee**, Jiwon Kim, Hojung Cha
 ACM International Conference on Mobile Computing and Networking (**MobiCom 2025**).
- [P6] Ember: Task Wakeup Sequence-Based Energy Optimization for Mobile Web Browsing
 Seonghoon Park, Jiwon Kim, **Jeho Lee**, Hojung Cha
 ACM SIGBED International Conference on Embedded Software (**EMSOFT 2025**).
- [P7] SecureRide: Detecting Safety-threatening Behavior of E-Scooters Using Battery Information
 Jiwon Kim, Geon Kim, **Jeho Lee**, Thiemo Voigt, Hojung Cha
 ACM SIGBED International Conference on Embedded Software (**EMSOFT 2025**).
- [P8] ARIA: Optimizing Vision Foundation Model Inference on Heterogeneous Mobile Processors for Augmented Reality
 Chanyoung Jung*, **Jeho Lee***, Gunjoong Kim, Jiwon Kim, Seonghoon Park, Hojung Cha
 ACM International Conference on Mobile Systems, Applications, and Services (**MobiSys 2025**).
Best Paper Award (2 of 233 submissions).

- [P9] Panopticus: Omnidirectional 3D Object Detection on Resource-constrained Edge Devices
Jeho Lee, Chanyoung Jung, Jiwon Kim, Hojung Cha
ACM International Conference on Mobile Computing and Networking (**MobiCom 2024**).
- [P10] Vulture: Cross-Device Web Experience with Fine-Grained Graphical User Interface Distribution
Seonghoon Park, **Jeho Lee**, Yonghun Choi, Hojung Cha
IEEE Conference on Computer Communications (**INFOCOM 2024**).
- [P11] OmniLive: Super-Resolution Enhanced 360° Video Live Streaming for Mobile Devices
Seonghoon Park*, Yeonwoo Cho*, Hyungchol Jun, **Jeho Lee**, Hojung Cha
ACM International Conference on Mobile Systems, Applications, and Services (**MobiSys 2023**).
- [P12] Crow API: Cross-device I/O Sharing in Web Applications
Seonghoon Park, **Jeho Lee**, Hojung Cha
IEEE Conference on Computer Communications (**INFOCOM 2023**).

Invited Articles, Posters, and Extended Abstracts

- [A1] 3DGS on Your Phone: Towards Fully Immersive Mobile Volumetric Video Streaming
Gunjoong Kim*, Seonghoon Park*, **Jeho Lee**, Chanyoung Jung, Hyungchol Jun, Hojung Cha
ACM GetMobile: Mobile Computing and Communications, 30(1), 2026. Invited article.
- [A2] VFM in Your Hands: Optimizing Real-Time Scene Understanding for Mobile Augmented Reality
Jeho Lee*, Chanyoung Jung*, Gunjoong Kim, Jiwon Kim, Seonghoon Park, Hojung Cha
ACM GetMobile: Mobile Computing and Communications, 29(4), 2025. Invited article.
- [A3] Towards Accurate, Adaptive, and Real-time Machine Perception on Resource-constrained Platforms
Jeho Lee
ACM MobiSys 2025 Rising Stars Forum.
- [A4] Poster: Mixture of Class-aware Experts for Efficient AIoT Inference
Hyemin Jeong, **Jeho Lee**, Seunghyeok Jeon, Hojung Cha
ACM MobiSys 2025 Poster.

RESEARCH PROJECTS

Development of On-device DNN Inference System for Real-time 3D Perception with Mobile 360-degree Camera

May 2024 – Present

National Research Foundation of Korea (NRF).

Research on efficient on-device inference for real-time 3D perception from mobile 360-degree cameras.

Task Relation Graph Prediction based on RNN

Mar 2023 – Feb 2024

Samsung Electronics, Republic of Korea.

Research on recurrent neural networks for task relation graph prediction.

PATENTS

- [T1] Method for Omnidirectional 3D Object Detection, Program Performing the Method, and Computing Device Executing the Program
Jeho Lee, Chanyoung Jung, Seonghoon Park, Hyungchol Jun, Hojung Cha
Republic of Korea Patent No. **10-2991405**, granted Jul 10, 2026.
- [T2] I/O Sharing Device and Method
Seonghoon Park, **Jeho Lee**, Hojung Cha
Republic of Korea Patent No. **10-2823808**, granted Jun 18, 2025.

TEACHING EXPERIENCE

Guest Lecturer, Ewha Womans University

Spring 2026

Towards Real-time Machine Perception on Resource-constrained Mobile Platforms

Guest Lecturer, Yonsei University

Spring 2024

On-Device Intelligence for 3D Perception: Challenges and Innovations

Teaching Assistant, Yonsei University
Operating Systems (CSI3101)

Spring 2023

Teaching Assistant, Yonsei University
Introduction to Computer Science (CSI2106)

Fall 2022

MENTORING EXPERIENCE

Dasol Yoon, Master's student, Yonsei University 2025 – Present
On-device AI with Mobile NPUs.

Jaehee Kim, Master's student, Yonsei University 2025 – Present
viNPU (EuroSys 2026).

Hyemin Jeong, Master's student, Yonsei University 2025
Mixture of Class-aware Experts for Efficient AIoT Inference (MobiSys 2025 Poster).

Gunjoong Kim, Master's student, Yonsei University 2024 – 2025
ARIA (MobiSys 2025) and Vega (MobiCom 2025).

Chanyoung Jung, Master's student, Yonsei University 2024 – 2025
ARIA (MobiSys 2025) and Vega (MobiCom 2025).

Chanyoung Jung, Undergraduate student, Yonsei University 2022 – 2023
Panopticus (MobiCom 2024).

Software Capstone Design, Yonsei University
Spring 2021, Fall 2021, Spring 2022, Spring 2023, Fall 2023.

SKILLS

Programming: Python, C/C++, Java.

ML Software: QNN, LiteRT (TensorFlow Lite), ONNX, AIMET, TensorRT, PyTorch, ExecuTorch.

Platforms: Qualcomm Snapdragon (Hexagon NPU), NVIDIA Jetson, Android, Linux.

Languages: English (fluent), Korean (native).

ACADEMIC SERVICE

- IEEE Transactions on Mobile Computing (TMC), Reviewer.
- IEEE Transactions on Networking (ToN), Reviewer.
- ACM MobiSys 2024, Student Volunteer.